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4TH INTERNATIONAL ONE HEALTH CONFERENCE -

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INTEGRAL ECOLOGY AS A LIGHT TO ADVANCE
THE ONE HEALTH VISION
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Domain 4: Innovation and Action

AI for Farmers: A YOLO and Language Model-Based Plant Disease Detection and Advisory System

SDSU | San Diego State
University



Launch Demo App



SPEAKER : AICHU TAN

AI-ε For Farmer Researchers



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One Health Integration

Healthy plants = Healthy people, animals, ecosystems

What is One Health?

Why Plant Health Matters

Recognizes the interconnected health of **humans, animals, plants,** and the **environment.**

Promotes collaboration across sectors for sustainable health solutions.

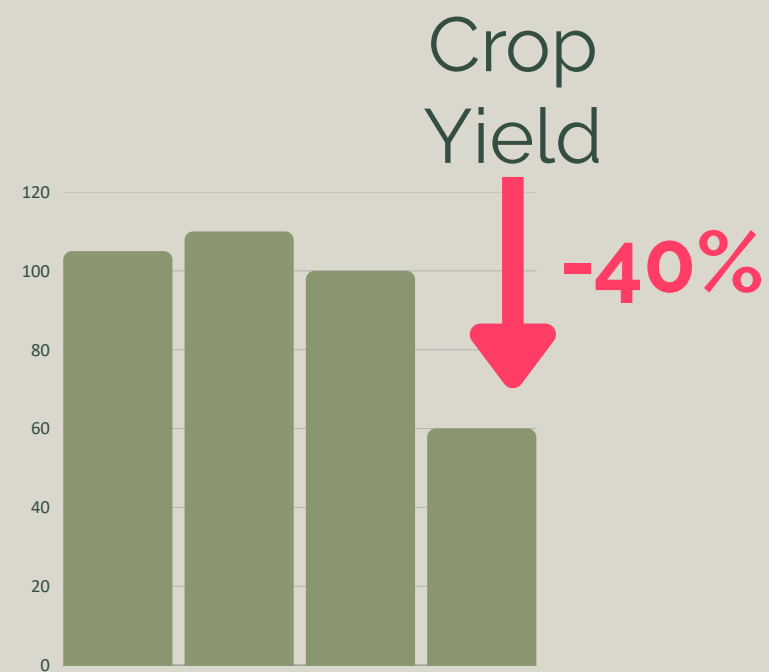
- **Food security** – protects yields and farmer livelihoods
- **Nutrition** – supports access to vitamins, proteins, and minerals
- **Animal health** – safe feed and reduced toxin/pesticide exposure
- **Ecosystems** – soil, water, and biodiversity resilience



**SUSTAINABLE
DEVELOPMENT** **GOALS**

The Challenge

- **FAO** estimates: up to **40%** of global **crop yields** are lost annually to pests and diseases [1]
- This is not just an agricultural problem. It's also a **health, nutrition, and climate problem.**
- These losses disproportionately affect **smallholder farmers**, driving the need for **timely, accessible solutions.**





Problem Statement

- **Smallholder farmers** most affected by crop losses.
- Limited access to extension services, costly lab tests, and difficult field diagnosis.
- **Overreliance on pesticides** → health risks, biodiversity loss, water contamination [21]
- Farmers often **struggle to distinguish diseases** in the field, leading to delays or inappropriate pesticide use



Research Gaps



- **Well-documented burden:** Up to 40% global crop yield loss due to pests & diseases, but few field-ready tools exist.
- **AI tools:** CNNs and YOLO models achieve high accuracy on benchmark datasets, yet lack real-world validation.
- **Mobile apps:** PlantVillage Nuru (cassava only, offline) and Plantix (multi-crop, online, commercial) show value, but face barriers:
 - Limited crop coverage
 - Internet dependence
 - Commercial restrictions

Study Objectives

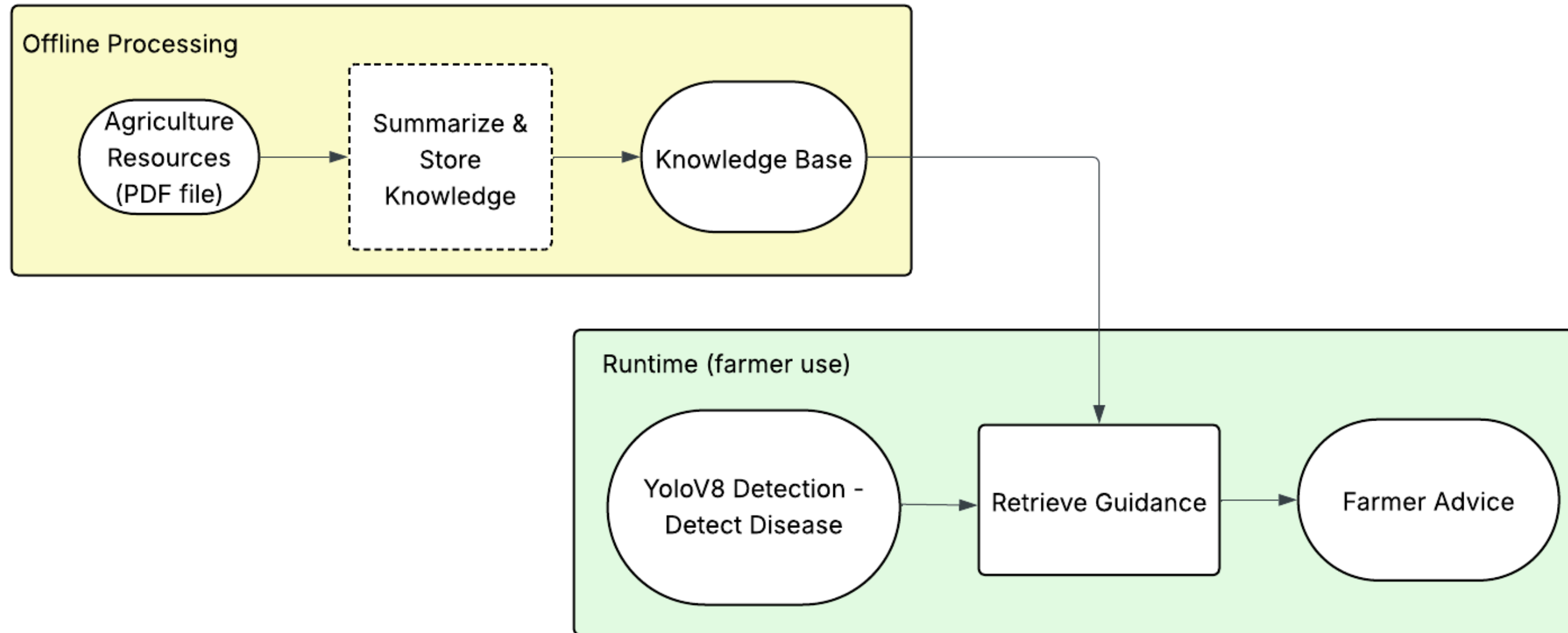
AI-ε: An open-source, offline-friendly prototype combining **YOLOv8** and **LangChain**.

- Detect diseases in Cashew, Cassava, Tomato, Maize
- Deliver safe, farmer-friendly prevention & treatment advice
- Empower smallholder farmers within a One Health framework





Methodology - Workflow



Farmer gets advice: **clear, safe, practical**



Methodology- Technical Details

01

Detection (YOLOv8m)

- **13,519 images** → 4 crops (cashew, cassava, maize, tomato) → 20 disease classes
 - Data split + augmentation (~**32k training samples**)
 - Bounding boxes created in Roboflow
-

02

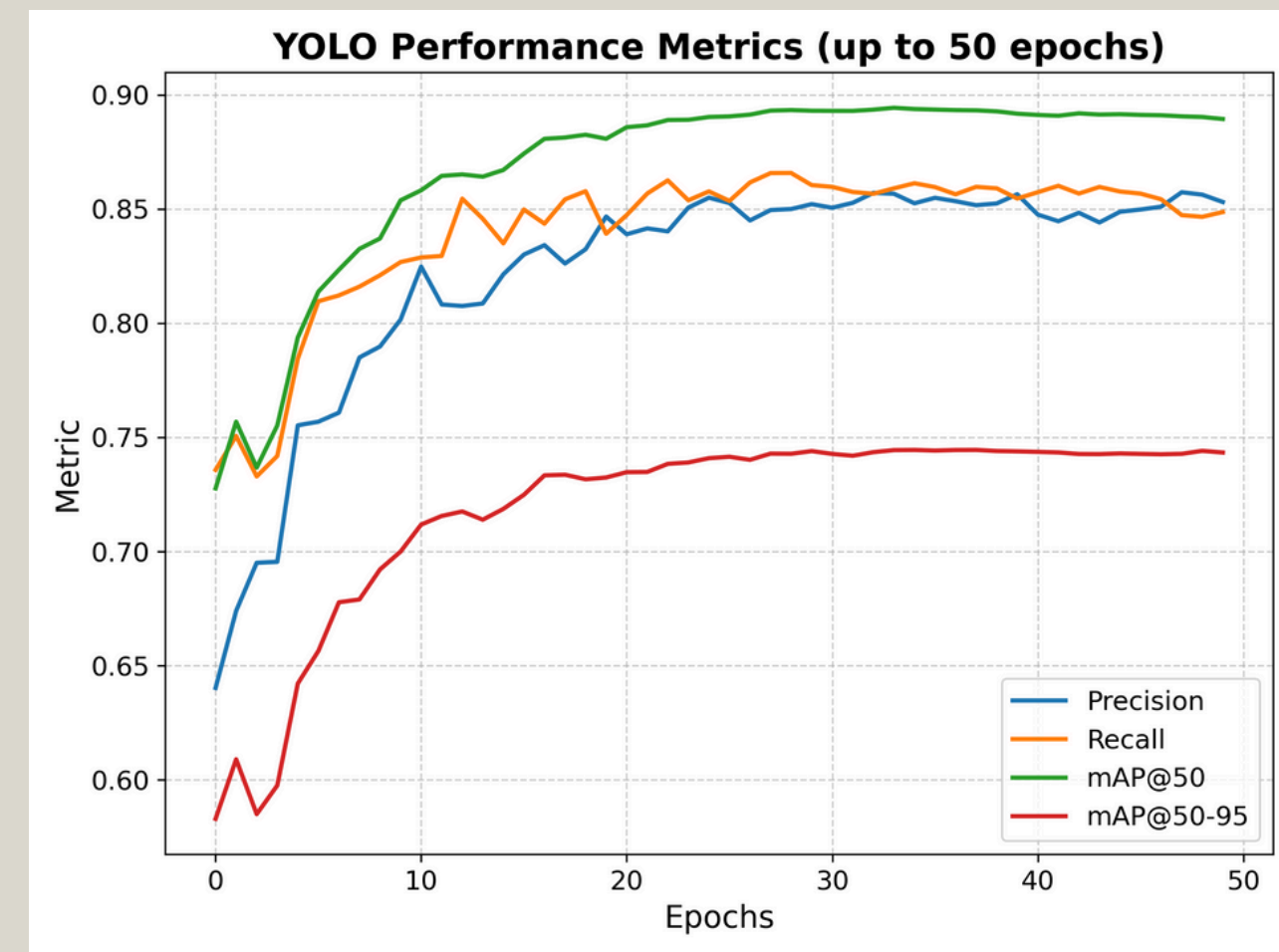
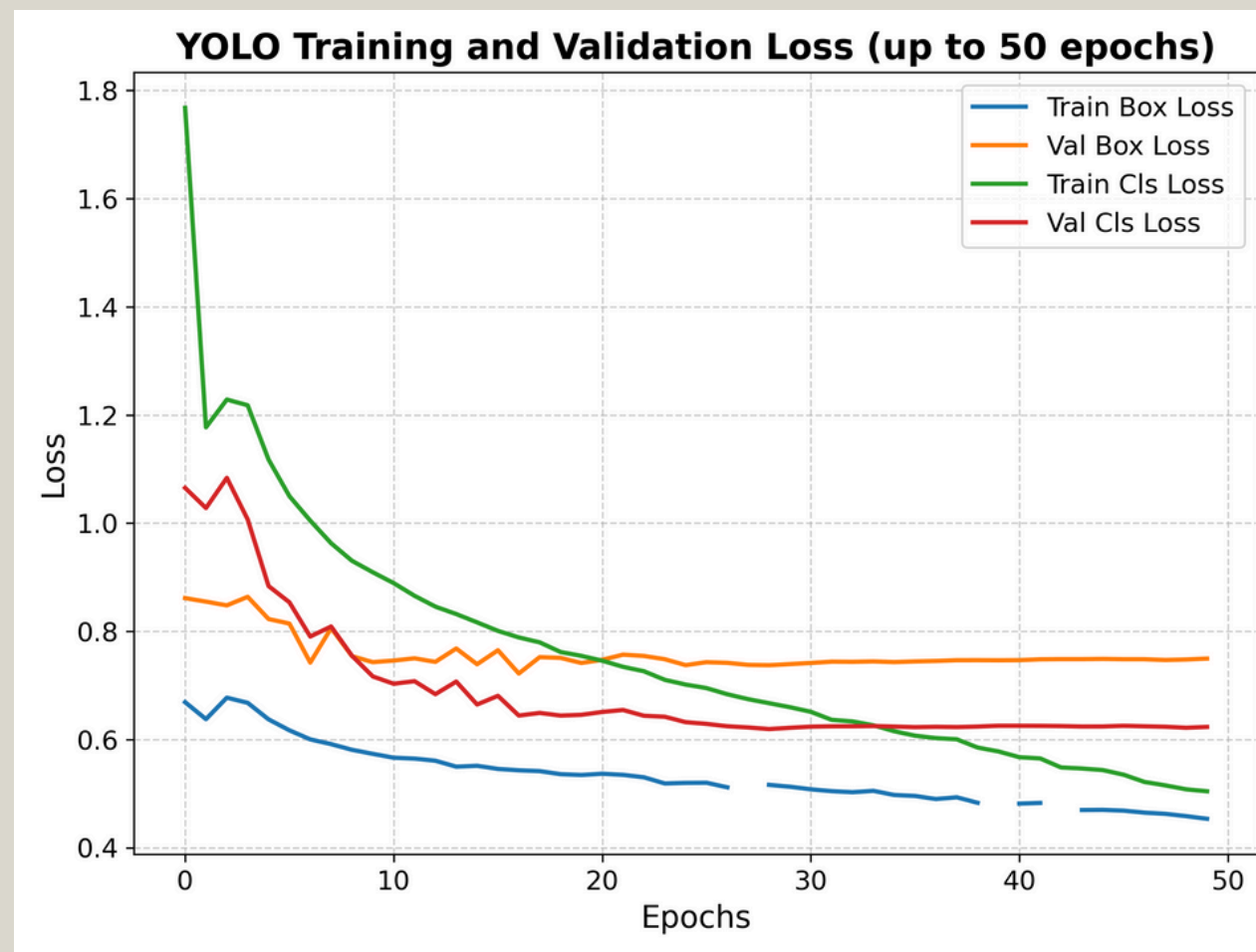
Advisory (LangChain RAG)

- Built from manuals + scientific reports
- Retrieval system delivers farmer-friendly prevention & treatment advice
- Non-chemical first; chemical only as last resort with PPE reminders

Together, these two phases create a **lightweight, offline-friendly** tool for **smallholder farmers**.

Results - Detection Performance (YOLOv8)

- Overall performance: **Precision 0.85, Recall 0.86, mAP@50 0.89**
- **Strongest:** Cassava mosaic, Cashew red rust, Cassava green mite
- **Weakest:** Cassava brown spot, Cassava bacterial blight, Maize leaf spot
- **Key insight:** Clear symptoms → higher accuracy; overlapping symptoms → lower accuracy



mAP = 0.89

P = 0.85

R = 0.86

Results - Detection Examples

- Strong detections with clear symptoms
- Lower accuracy with overlapping or subtle symptoms
- Model generalizes across crops but challenges remain

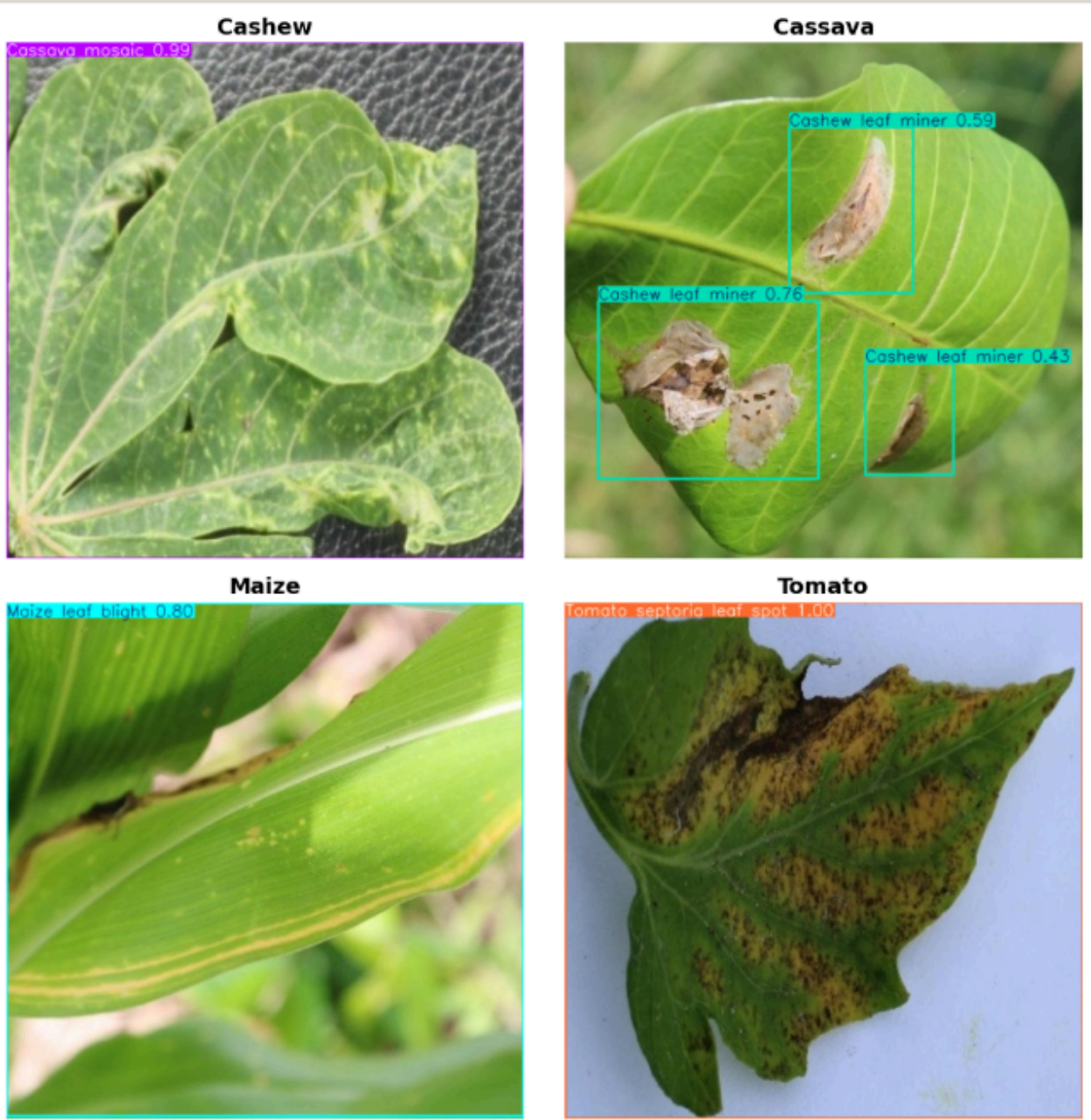


Table 4.1 Class-wise performance of YOLOv8 at epoch 50, showing precision, recall, mAP@50, mAP@50-95, and F1 score for selected crop-disease classes.

Crop / Disease	Precision (P)	Recall (R)	mAP@50	mAP@50-95	F1
Overall (all classes)	0.852	0.862	0.894	0.745	0.857
Cassava mosaic	0.989	1	0.995	0.98	0.994
Cashew red rust	0.976	0.981	0.989	0.97	0.978
Cassava green mite	0.987	0.907	0.987	0.97	0.945
Cassava brown spot	0.669	0.529	0.611	0.24	0.591
Cassava bacterial blight	0.709	0.695	0.722	0.348	0.702
Maize leaf spot	0.749	0.52	0.617	0.411	0.614



Advisory Output

```
=== Tomato - Septoria Leaf Spot ===  
=== Prevention ===  
- Plant resistant tomato varieties to Septoria leaf spot.  
- Space tomato plants properly for good air circulation.  
- Remove and destroy infected leaves as soon as they are noticed.  
- Mulch around tomato plants to prevent soil splashing onto leaves.  
- Rotate tomato crops with non-related plants to break the disease cycle.  
- Inspect tomato plants weekly for early signs of Septoria leaf spot.  
  
=== Treatment ===  
Non-chemical:  
- Remove infected leaves and stems promptly.  
- Ensure good spacing between plants for airflow.  
  
Chemical (last option):  
- Copper fungicide – when: signs of Septoria leaf spot appear; interval:  
every 7-10 days; note: Follow label instructions and wear PPE.  
  
Seek help when:  
- Disease continues to spread despite treatment efforts.
```

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Results: Advisory Module

Principle: Validated against trusted literature.

- Prioritizes non-chemical measures: pruning, sanitation, resistant varieties.
- Chemicals only as a last resort, with PPE reminders.

Example – Tomato Septoria Leaf Spot

- Prevention: resistant varieties, pruning, crop rotation.
- Treatment: copper fungicide only if necessary.

Advisory system delivers **safe, farmer-friendly, and sustainable guidance.**



One Health Contributions

Section 1 – Health

- **Human health** → less pesticide exposure, safer food
- **Animal health** → cleaner feed, less chemical runoff
- **Environmental health** → pollinator & water protection

Section 2 – Food & Equity

- **Nutrition** → more stable yields for food security
- **Equity** → open-source, offline access for smallholders

Section 3 – Alignment

- Supports FAO Quadripartite **One Health vision**



Limitations

- **Scope:**

Prototype supports only 4 crops.

- **Accuracy:**

Struggles with diseases that present subtle or overlapping symptoms

- **Advisory:**

Retrieval-based, not yet conversational or interactive

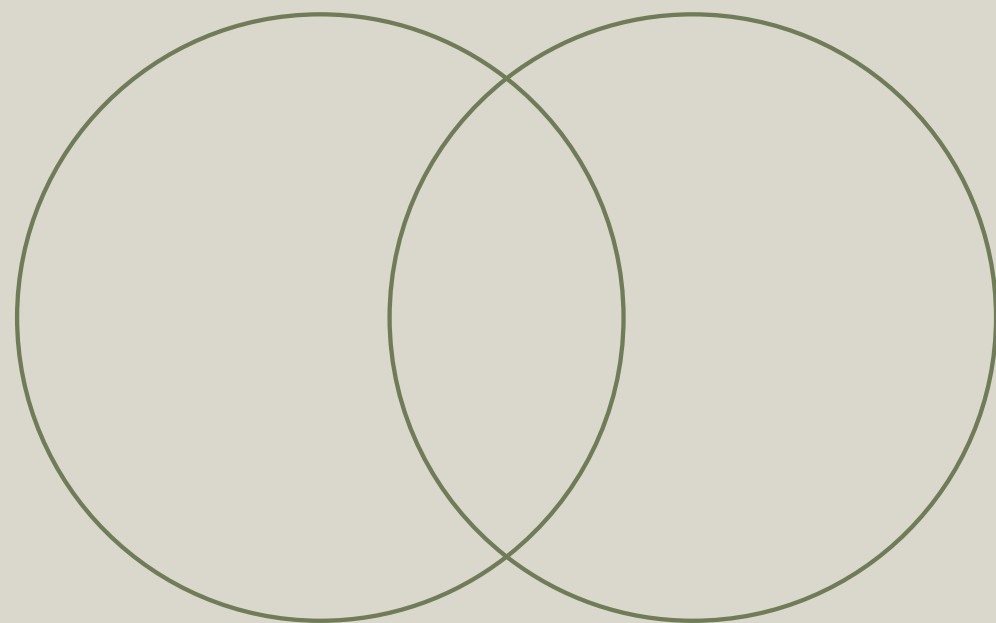
- **Accessibility:**

No multilingual or low-literacy features.

These limitations highlight areas for future development.



Future Research



Expand coverage

- Add more staple & region-specific crops

Improve accuracy

- Larger, more diverse datasets
- Explore YOLOv11 and advanced models

Enhance advisory system

- Move to interactive, conversational design
- Add context-aware, personalized recommendations

Increase accessibility

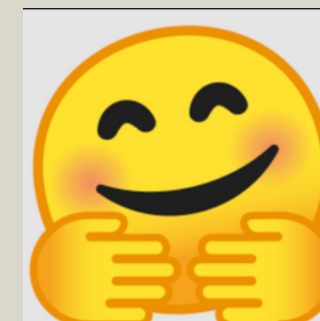
- Multilingual support
- Low-literacy features (icons, audio guidance)
- Offline-first deployment for rural areas

Field validation & feedback

- Pilot testing with smallholder farmers
- Iterative improvements based on real-world use

Acknowledgements

- We thank the **Metabolism of Cities Living Lab, San Diego State University**, for their support and collaboration.
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- Full list available on request.

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Thank You!



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